DNA Structure

Exam code: 0610 & 0970

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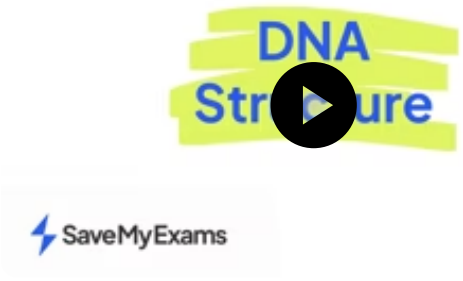

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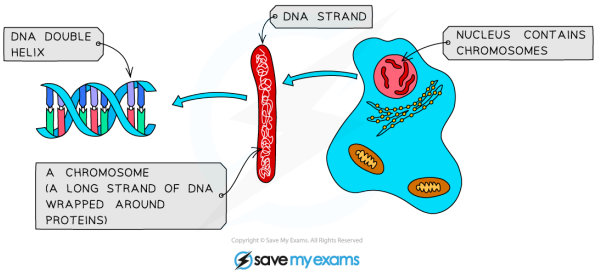
Help me understand

Structure of a DNA Molecule: Extended

Extended Tier Only

- DNA, or deoxyribonucleic acid, is the molecule that contains the instructions for the growth and development of all organisms
- It consists of two strands of DNA wound around each other in what is called a **double helix**



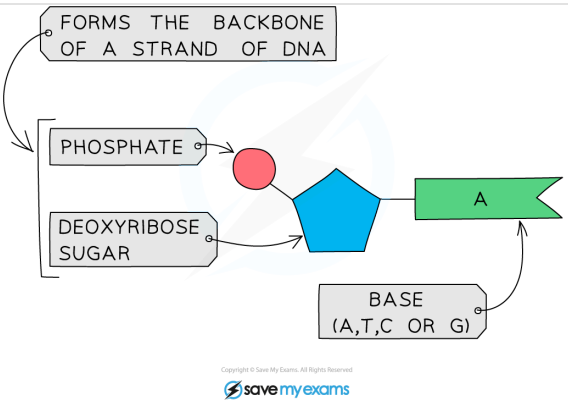
DNA, chromosomes and the nucleus

- The individual units of DNA are called **nucleotides**

Stuck on something?

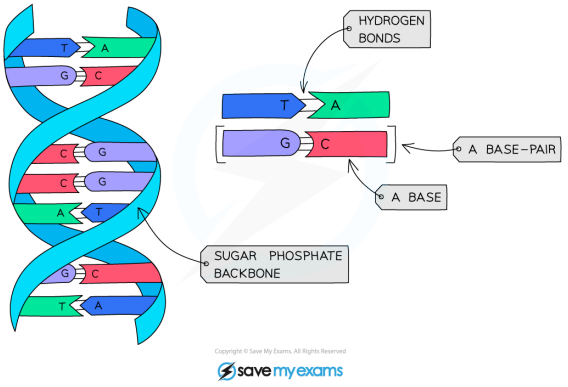
Ask questions and get help with exactly what you're revising.

Got it



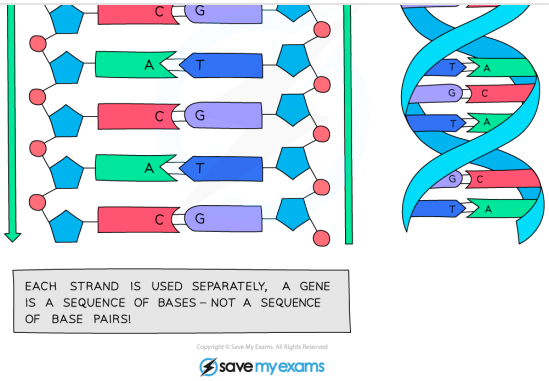
A nucleotide

- All nucleotides contain the same phosphate and deoxyribose sugar, but differ from each other in the **base** attached
- There are four different bases, **Adenine (A), Cytosine (C), Thymine (T) and Guanine (G)**
- The bases on each strand pair up with each other, holding the two strands of DNA in the double helix
- The bases always pair up in the same way:
  - Adenine always pairs with Thymine **(A-T)**
  - Cytosine always pairs with Guanine **(C-G)**



DNA base pairs

- The phosphate and sugar section of the nucleotides form the 'backbone' of the DNA strand (like the sides of a ladder) and the base pairs of each strand connect to form the rungs of the ladder



The DNA helix is made from two strands of DNA held together by hydrogen bonds

- It is this sequence of bases that holds the code for the formation of proteins

Examiner Tips and Tricks

You do not need to learn the names of the bases, **just their letter**. Make sure you **know which bonds with which**, as this is the most commonly asked question about this topic.

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